

Appendix K Analysis Plan

K.1 Scope

This document outlines how the scores produced by Phase 2 and Phase 4 are to be analysed, and it does so at the level of the decision rather than at the level of the finding.

K.2 What is analysed

All scores analysed are held in the scoring workbook at Appendix I. A requirement-area score is one rater's mark out of 100 for one requirement area of one policy.

Sixteen model scores: Four cross-audits, A(P1a), A(P1b), A(P2a) and A(P2b), each scoring Requirements 7, 8, 10 and 12.

Forty-eight human scores: R1, R2 and R3, each scoring four policies across the same four areas.

Those sixty-four are the analysis set. The sixteen policy-level scores are averages of them, reported but not tested, for the reason at K.3.

A coverage figure is built from an item-by-item checklist. Only Requirement 8's was transcribed into the workbook, where twenty-four sub-requirements apply, so its coverage alone can be traced back to the ratings behind it. The limitation is recorded rather than passed over.

K.3 Decisions taken before the analysis

Unit of analysis. The requirement-area score is the unit. Policy-level scores are reported as descriptive figures but carry no agreement statistic: taking them as the unit would reduce the model-to-human comparison to four paired values, and a coefficient on four pairs would carry an interval too wide to inform anything.

K.3.1 The coverage denominator

Coverage is calculated against the applicable sub-requirement counts fixed in the requirement extract at Appendix C: nine for Requirement 7, twenty-four for Requirement 8, nine for Requirement 10 and thirty for Requirement 12, seventy-two in total. No other denominator is used, and the seventy-two is never the denominator for a single area. Where an auditor returned a coverage figure that

cannot be reconciled with those counts, the figure is reported as returned and the discrepancy described in Chapter 4. A returned score is never quietly recomputed.

Two such discrepancies exist, and they are named here so that Chapter 4 cannot pass over them. Both fall in A(P2b), the ChatGPT audit of the Grok compliance-framing policy. For Requirement 7 the checklist total is 22 points of a possible 27, which gives a calculated coverage of 81, yet the auditor returned 92; for Requirement 12 it is 74 of a possible 90, which gives 82, yet the auditor returned 85. The other fourteen requirement-area cells across the four audits reconcile exactly. Both are reported as returned, with the calculated figure shown beside the returned one and the gap stated. Neither is replaced, because a coverage figure that does not follow from the auditor's own checklist is itself evidence about how the model applied the instrument.

K.3.2 Findings charged to the wrong criterion

Both model auditors were observed to reduce coverage where the policy stated an obligation correctly but showed no evidence of implementation. Under Appendix D, Section D.3, an implementation gap belongs to proportionality, not to coverage. Every score is reported exactly as the auditor returned it, and the misallocation is treated as a finding about auditor behaviour rather than an error to repair. A recomputed figure may be shown once, clearly labelled, to show the size of the effect. It enters no table of results, and no statistic is calculated from it.

Silently repairing the scores was considered and rejected. Doing so would have substituted what the researcher wished had happened for what was being measured, which is whether a language model applies a rubric as written. It is the misallocation itself, not the score a correct allocation would have produced, which bears on Objective 2.

K.3.3 Severity and rounding

Severity. Findings carry the severity the source assigned, and none is re-graded. Where a source left severity blank, the finding is counted in any total but excluded from any severity distribution, and the number excluded is stated wherever a distribution is reported.

Rounding. Rounding occurs once, when the requirement-area score is formed, to the nearest whole number, as the workbook formula does. Means of rounded

scores are reported to one decimal place. No intermediate value is rounded, and no rounded value is rounded again.

K.3.4 Choice of agreement statistic

Agreement between the model auditor and the human panel is reported using Gwet's AC1 (Gwet, 2008) rather than Cohen's kappa (Cohen, 1960). The reason is the shape of the data, not a preference between authors. R1, R2 and R3 cluster tightly in the upper bands. Where the marginal distribution is skewed in that way, kappa can return a low coefficient even alongside very high observed agreement, which is the first of the two paradoxes set out by Feinstein and Cicchetti (1990). AC1 uses a chance-agreement term that does not collapse under high prevalence. On that ground it appears the more informative statistic here.

The coefficient applies only to the three judged criteria: accuracy, proportionality and enforceability. Appendix D, Section D.4, instructs raters to enter a band midpoint of 90, 70, 50, 30 or 10, but several entered intermediate values such as 95, 85 and 65, and one model auditor entered 100. Every judged score is mapped back to its band before the coefficient is calculated, using the rubric's own boundaries. The mapping changes no recorded score; raw entries stay in the workbook and are what the descriptive tables report. All five bands form the category space whether or not all five are used. Coverage is continuous, so no coefficient of this family applies to it, and coverage agreement is reported as the mean absolute difference and the largest single absolute difference, with the direction stated.

Percentage agreement accompanies AC1 throughout, so a reader who prefers the plainer figure has it. Chapter 3 §3.12 names Cohen's kappa conditionally, reporting a coefficient "only where the number of reviewers makes it meaningful"; that wording stands, and the substitution is declared here rather than left to be discovered. AC1 is calculated pairwise, the model auditor against each reviewer in turn, with the three coefficients reported alongside their mean; agreement within the panel is reported separately as a three-rater coefficient.

K.4 The three comparisons

Three comparisons are drawn. Figure K.1 shows how the policies were allocated to auditors and reviewers, and which scores each comparison uses.

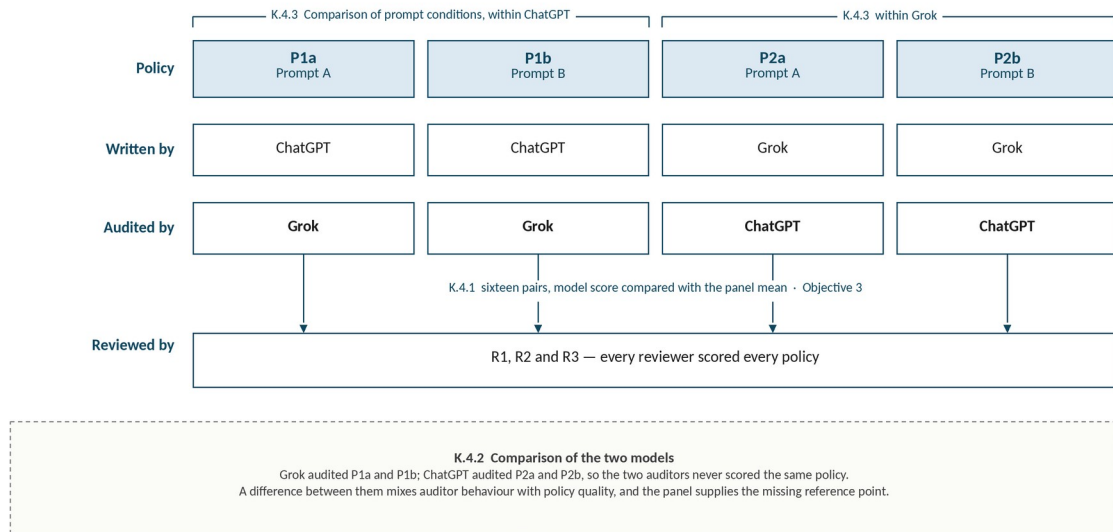


Figure K.1: Allocation of policies to auditors and reviewers.

K.4.1 Comparison of model and human scores

Sixteen pairs are formed, one per policy and requirement area, setting the model auditor's score against the mean of R1, R2 and R3 for the same cell. Reported are the signed difference, which shows systematic leniency or severity, and the absolute difference, which shows dispersion irrespective of direction, each per pair and as a mean across the set. AC1 is calculated on the banded criteria as described at K.3.4. This comparison carries Objective 3, and it is the only one of the three with a human standard against which model behaviour can be judged.

K.4.2 Comparison of the two models

The two auditors never scored the same policy, as Figure K.1 shows, so a difference between their scores confounds auditor behaviour with policy quality and cannot on its own establish which model is the stricter auditor. The comparison is reported because the divergence is large, with the confound stated alongside it. Where the panel and one model agree closely on a policy the other scored very differently, that triangulation is the evidence; the raw model-to-model difference is not.

K.4.3 Comparison of prompt conditions

Standard-reference (Prompt A) is compared against compliance-framing (Prompt B) within each generating model, giving two within-model comparisons rather than

one pooled figure. Pooling was rejected because the models differ in level as well as in response to the prompt, and a pooled figure would mix the two effects. The comparison speaks to Objective 1. With one policy per cell it supports description of the direction and size of any prompt effect, and nothing stronger.

K.5 Answering Objective 2 without a refinement phase

Phase 3 was not run. Appendix B and the approved Project Description Form both describe refinement as a contingency rather than a fixed stage, so the omission is not a deviation and is not recorded as one. It does remove the before-and-after measurement the second half of Objective 2 anticipated, and the analysis answers for that rather than passing over it.

Objective 2 has two limbs. The first asks whether a model can identify compliance gaps in another model's policy. Phase 2 and the comparison at K.4.1 answer it in full: the audits exist, their findings are recorded, and R1, R2 and R3 provide the standard against which their accuracy is assessed. The second limb, whether audit feedback drives measurable improvement in coverage, is answered by argument from the findings rather than by measurement of a revised policy. Two grounds carry that argument, both recorded before the scores were examined.

What the analysis reports in place of a measured improvement is the proportion of each auditor's findings that are actionable within the document itself, classified against the criterion the finding names. That figure bears directly on the practical question the second limb asks, which is whether audit feedback from a language model is the kind of feedback a policy author could act on. It is a weaker answer than a measured delta. It is presented as such.

K.6 What this plan does not permit

The following are excluded, and the exclusions hold even where the excluded analysis would be more favourable to the study's conclusions.

No significance test is reported on any comparison. Sixteen model scores and three reviewers do not support one, and a p-value calculated on this sample would carry an authority the design cannot bear.

No score is adjusted, reweighted or recomputed after the fact, save for the single labelled illustration permitted at K.3.2.

No requirement area is dropped from a comparison because it behaves differently from the others. Requirement 10 in particular is expected to score differently, for the reason Chapter 3 §3.15 already gives, and it stays in.

No reviewer is identified other than by code. R1, R2 and R3 are used throughout, in the workbook, in the chapters and in the appendices, as the participant documents at Appendix G require.

No claim is made about models in general. Every finding is stated of the two builds named in the run details, on the dates recorded there.